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Passive Gravity Gradient Capture During In-Space Assembly and Manufacturing

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In-space assembly and manufacturing (ISAM) can overcome the volume limitations of rocket fairings and enable the construction of large structures optimized for the space environment. For ISAM of structures with dimensions larger than approximately 100 m, our previous work has highlighted how the capabilities of the ISAM spacecraft's attitude control system (ACS) result in minimum achievable fabrication times ranging from months to years. This paper explains how to use gravity gradient torques during the ISAM process to overcome these ACS limitations and passively capture the ISAM structure and spacecraft into a gravity-gradient-stabilized final orientation after construction. Our concept, referred to as passive gravity gradient capture, is illustrated during ISAM of two representative truss structures—a 2D triangle unit cell and a 3D curved gridshell—in circular Earth orbits. Each case study uses the time-reversibility of the underlying equations of motion to compute initial conditions for passive attitude trajectories that terminate in stable final orientations. Monte Carlo simulations show that these passive attitude trajectories are sensitive to initial conditions; however, alternate strategies such as deploying a gravity gradient boom before construction can enable robust gravity gradient capture of large structures. Taken together, the results demonstrate the feasibility of passive gravity gradient capture during ISAM and motivate further development of strategies that exploit in-space environmental disturbances for attitude control during ISAM.

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Nomenclature

A	=	strut cross-sectional area, m^2
A_C	=	disk cross-sectional area, m^2
C_D	=	drag coefficient
D	=	diameter, m
E	=	total energy, J
$\hat{\boldsymbol{e}}$	=	random unit vector
$\boldsymbol{F}$	=	applied force, N
F/D	=	focal-length-to-diameter ratio
$\boldsymbol{h}_{\mathrm{CM}}$	=	relative angular momentum, $kg \cdot m^2 \cdot s^{-1}$
$[\boldsymbol{I}_{\mathrm{CM}}]$	=	mass moment of inertia matrix about center of mass, $kg \cdot m^2$
L	=	strut length, m
M	=	total mass, kg
m_i	=	point mass, kg
m_{SC}	=	instantaneous spacecraft mass, kg
m_{SC0}	=	final spacecraft mass after construction, kg
m_T	=	instantaneous truss mass, kg
m_{tip}	=	tip mass, kg
m_{T0}	=	final truss mass after construction, kg
N	=	total number of struts
n	=	orbit mean motion, $rad \cdot s^{-1}$
$\hat{\boldsymbol{n}}$	=	unit surface normal vector
P	=	solar radiation pressure, $N \cdot m^{-2}$
$\boldsymbol{q}$	=	unit quaternion
$\hat{\boldsymbol{R}}$	=	orbit radius unit vector
$\boldsymbol{r}_{\mathrm{CM}}$	=	position of center of mass relative to spacecraft body frame, m
$\boldsymbol{r}_i$	=	position of mass m_i relative to spacecraft body frame, m
s	=	spacecraft side length, m
T	=	orbit period, s
t	=	time, s
$\hat{t}$	=	nondimensional time (normalized by orbit period)
$\hat{t}_F$	=	nondimensional construction timescale (normalized by orbit period)
$\boldsymbol{v}$	=	velocity vector, $m \cdot s^{-1}$
$\boldsymbol{\alpha}$	=	angular acceleration, $rad \cdot s^{-2}$
Γ	=	strut extrusion profile, m
γ_s	=	sun angle, rad
$\boldsymbol{\delta}$	=	offset between center of pressure and center of mass, m
$\delta\theta$	=	angle perturbation, rad
$\delta\hat{\omega}$	=	nondimensional angular velocity perturbation
ϵ	=	disk height-to-diameter ratio
η_I	=	truss-to-spacecraft inertia ratio
η_M	=	truss-to-spacecraft mass ratio
θ_z	=	pitch angle, rad
κ_T	=	disk offset scale factor
μ_E	=	Earth gravitational parameter, $m^3 \cdot s^{-2}$
ρ	=	density, $kg \cdot m^{-3}$
ρ_A	=	atmospheric density, $kg \cdot m^{-3}$
$\boldsymbol{\tau}$	=	applied torque, N·m
ϕ	=	tilt angle, rad
$\boldsymbol{\omega}$	=	angular velocity, $rad \cdot s^{-1}$
$\hat{\boldsymbol{\omega}}$	=	nondimensional angular velocity

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I. Introduction

IN-SPACE assembly and manufacturing (ISAM) can transform the capabilities of modern space systems by overcoming the volume limitations of rocket fairings. In doing so, ISAM can enable the construction of structures that are optimized for loads in the space environment and substantially larger than those achievable with modern deployable structure technologies [1,2]. The construction of

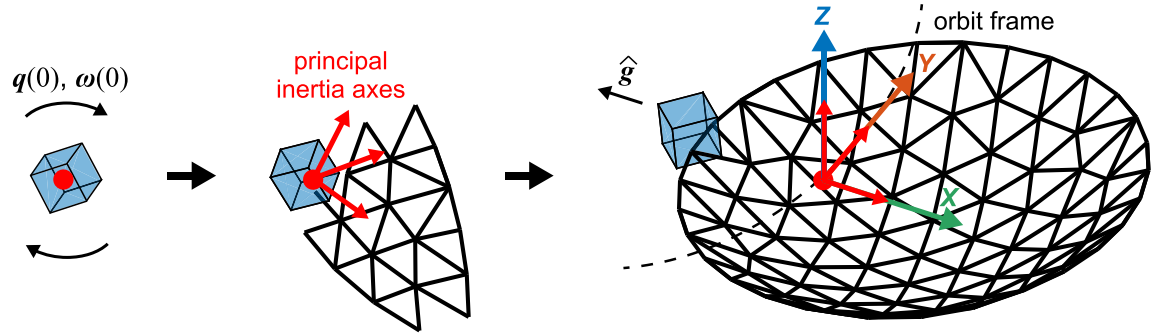


Fig. 1 Gravity gradient capture for ISAM: a passive attitude trajectory from a nonzero initial condition yields a gravity-gradient-stabilized final orientation that aligns the principal inertia axes with the orbit frame.

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truss support structures, in particular, has been cited as an enabling technology for concepts ranging from next-generation space telescopes [3] to high-power communication systems [4,5] and space-based solar power [6]. Numerous ISAM concepts are currently under development, including the robotic assembly of repeated unit cells [7–9], the extrusion of carbon fiber-reinforced thermoplastics [10–12], and the deformation processing of ductile rods [13,14].

An important consideration for ISAM spacecraft is attitude control system (ACS) design. Our previous analysis [15] showed that with the capabilities of modern spacecraft and proposed ISAM processes, the dominant constraint on the fabrication times of truss structures larger than approximately 100 m in size is the attitude control authority required to maintain a fixed inertial orientation. In the absence of environmental disturbance torques, the analysis estimated that the ACS capabilities resulted in fabrication times ranging from months to years. Including environmental disturbance torques in the analysis would have further increased these fabrication times.

At the same time, environmental disturbance torques provide opportunities for passive attitude control and stabilization. For example, gravity gradient torques have previously been used to stabilize the attitude of high-aspect-ratio spacecraft such as NASA's Long Duration Exposure Facility [16] and to desaturate the control moment gyroscopes on NASA's Skylab space station [17]. Solar radiation pressure torques have also been used for attitude control and angular momentum management on multiple past spacecraft, including NASA's Mariner [18], NASA's MESSENGER [19], and the Japan Aerospace Exploration Agency's Hayabusa2 [20]. Similarly, the International Space Station (ISS) uses torque equilibrium attitudes [21] to negate the effects of environmental disturbances and reduce angular momentum storage requirements. The ISS also plans large-angle slew maneuvers that take advantage of environmental disturbances to avoid saturation of its control moment gyroscopes [22]. For ISAM spacecraft that undergo large changes in mass properties over long time horizons, similar strategies can reduce attitude control requirements during construction.

This paper assesses the feasibility of using gravity gradient torques to passively control the attitude of an ISAM spacecraft during the construction of large truss structures. Our approach, referred to as passive gravity gradient capture (Fig. 1), combines the attitude dynamics of the ISAM process with the gravity gradient torque to plan a passive attitude trajectory that terminates in a gravity-gradient-stabilized orientation. The use of passive gravity gradient capture eliminates any requirement for active attitude control during construction and prevents tumbling after construction is complete. Passive gravity gradient capture was previously demonstrated in lunar orbit in the 1970s during the deployment of 200-m-long, symmetric booms on NASA's twin Radio Astronomy Explorer (RAE) spacecraft [23–25]. Despite the successes of NASA's RAE spacecraft, the concept of passive gravity gradient capture is unexplored in the context of ISAM, particularly for ISAM processes that involve complex inertia changes over long timescales.

With this in mind, the objective of this paper is to characterize the feasibility of passive gravity gradient capture during the ISAM of truss structures. To that end, Sec. II first uses a scaling analysis to understand the orbital regimes where passive gravity gradient capture during ISAM is likely feasible. Section III then presents a model for attitude dynamics during ISAM, which is subsequently used in Sec. IV to illustrate passive gravity gradient capture during the construction of two representative truss structures: a 2D triangle unit cell and a 3D curved gridshell. In each case study, the ISAM spacecraft sequentially extrudes struts to assemble the truss structure, an approach that is representative of a range of candidate ISAM processes [8,9,14]. These case studies inform the discussion of strategies for robust gravity gradient capture in Sec. V. Finally, Sec. VI concludes by summarizing the main results of the paper.

II. Disturbance Scaling Analysis

This section presents a scaling analysis to illustrate the relative importance of the time-varying mass properties and various environmental disturbance torques on the attitude dynamics during ISAM in circular orbits. The analysis largely follows [26] but, importantly, includes an additional torque from the construction rate of the ISAM process.

Consider a thin cylindrical disk (Fig. 2) with constant density ρ, constant height-to-diameter ratio $\epsilon \ll 1$, and time-varying diameter $D(t)$ as a simplified model for ISAM of a circular antenna. The monotonically increasing diameter of the disk models the construction of the antenna aperture.

The attitude dynamics of the disk are modeled with the Newton–Euler equations of motion for a rigid body [27]:

$$[\boldsymbol{I}_{\mathrm{CM}}]\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times [\boldsymbol{I}_{\mathrm{CM}}]\boldsymbol{\omega} = \boldsymbol{\tau}_{\mathrm{MD}} + \boldsymbol{\tau}_{\mathrm{GG}} + \boldsymbol{\tau}_{\mathrm{AD}} + \boldsymbol{\tau}_{\mathrm{SRP}} \tag{1}$$

Here, the dot notation denotes differentiation with respect to time t, $\boldsymbol{\omega}$ is the angular velocity of the disk with respect to the inertial frame, and $[\boldsymbol{I}_{\mathrm{CM}}(t)]$ is the time-varying symmetric positive definite mass moment of inertia matrix about the center of mass given by

$$[\boldsymbol{I}_{\mathrm{CM}}(t)] = \frac{\pi}{32}\rho\epsilon D(t)^5 \mathrm{diag}\left\{\frac{1}{6}(3+4\epsilon^2), \frac{1}{6}(3+4\epsilon^2), 1\right\} \tag{2}$$

The terms on the right side of Eq. (1) are the torques from the time-varying mass distribution (MD) and environmental disturbances due to gravity gradients (GG), atmospheric drag (AD), and solar radiation pressure (SRP). Equation (2) only considers the mass properties

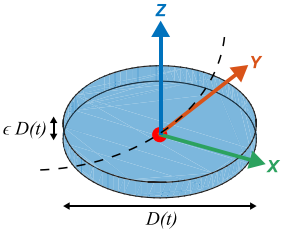


Fig. 2 Thin cylindrical disk with increasing diameter $D(t)$.

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of the disk; i.e., it neglects the mass properties of the ISAM spacecraft.

The torque due to the time rate of change of the inertia of the disk is [27]

$$\boldsymbol{\tau}_{\mathrm{MD}} = -[\dot{\boldsymbol{I}}_{\mathrm{CM}}]\boldsymbol{\omega} \tag{3}$$

The gravity gradient torque, which results from variations in the gravitational acceleration at different points in the disk, is [27]

$$\boldsymbol{\tau}_{\mathrm{GG}} = 3n^2(\hat{\boldsymbol{R}} \times [\boldsymbol{I}_{\mathrm{CM}}]\hat{\boldsymbol{R}}) \tag{4}$$

where $n = 2\pi/T$ is the mean motion of a circular orbit with period T, and $\hat{\boldsymbol{R}}$ is the orbit radius unit vector pointing from the center of the Earth to the center of mass of the spacecraft. The atmospheric drag and solar radiation pressure disturbances, respectively, result from collisions with atmospheric particles and photons from solar flux. These disturbances apply forces to the spacecraft, and if the center of pressure is offset from the center of mass, torques of the form $\boldsymbol{\tau} = \boldsymbol{\delta} \times \boldsymbol{F}$, where $\boldsymbol{\delta}$ is the vector from the center of mass to the center of pressure and $\boldsymbol{F}$ is the disturbance force. The disturbance forces from atmospheric drag and solar radiation pressure are [27]

$$\boldsymbol{F}_{\mathrm{AD}} = -\frac{1}{2}\rho_A C_D A_C v^2 \hat{\boldsymbol{v}} \tag{5}$$

$$\boldsymbol{F}_{\mathrm{SRP}} = 2PA_C \cos^2\gamma_s \hat{\boldsymbol{n}} \tag{6}$$

where ρ_A is the atmospheric density at the orbit altitude, C_D is the drag coefficient, A_C is the cross-sectional area of the disk, $\boldsymbol{v}$ is the orbital velocity vector with magnitude v and direction $\hat{\boldsymbol{v}}$, P is the solar radiation pressure, and γ_s is the angle between the surface normal vector $\hat{\boldsymbol{n}}$ and the sun. For simplicity, Eq. (5) does not account for the rotation rate of Earth's atmosphere.

The derivation of scaling relations for the angular accelerations due to these disturbance torques makes several simplifying assumptions similar to those in the previous analysis [26]. First, the analysis assumes that the disk spins slowly about its maximum principal inertia axis with an angular velocity that is approximately equal to the orbit mean motion (i.e., $\boldsymbol{\omega} \approx [0, 0, n]^T$). This is a reasonable assumption for nadir-pointing spacecraft and simplifies Eq. (1) so that the angular acceleration $\alpha \equiv \dot{\omega}$ is proportional to the moment of inertia (i.e., $\alpha = \tau/I$). Next, the analysis assumes that $\boldsymbol{\delta}$ remains in the plane of the disk (due to the small thickness of the disk) and that its magnitude is proportional to the instantaneous disk diameter, i.e., $\delta = \kappa_T D(t)$, where $\kappa_T > 0$ is a scale factor. To estimate the maximum angular accelerations, the analysis considers the cases where the cross-sectional area of the disk is perpendicular to the orbital velocity and sun vectors, as well as orientations that maximize the gravity gradient torque. With these assumptions, the magnitudes of the angular accelerations corresponding to each disturbance torque are

$$\alpha_{\mathrm{MD}} = 5n\dot{D}D^{-1} \tag{7}$$

$$\alpha_{\mathrm{GG}} = \frac{3n^2(3-4\epsilon^2)}{2(3+4\epsilon^2)} \tag{8}$$

$$\alpha_{\mathrm{AD}} = \frac{48\rho_A\kappa_T(n\mu_E)^{2/3}}{\rho\epsilon(3+4\epsilon^2)}D^{-2} \tag{9}$$

$$\alpha_{\mathrm{SRP}} = \frac{96\kappa_T P}{\rho\epsilon(3+4\epsilon^2)}D^{-2} \tag{10}$$

Equations (7–10) reveal several insights into attitude dynamics during ISAM. For example, Eq. (7) shows that the angular acceleration α_{MD} from the rate of change of inertia scales with the ratio of the diameter growth rate $\dot{D}$ to the instantaneous diameter D. As the gravity gradient torque [Eq. (4)] is proportional to the inertia, Eq. (8) shows that the angular acceleration α_{GG} due to gravity gradients is independent of size but proportional to n^2. In contrast (and perhaps counterintuitively), Eqs. (9) and (10) demonstrate that the angular accelerations α_{AD} due to atmospheric drag and α_{SRP} due to solar radiation pressure scale with D^{-2}. This occurs because the atmospheric drag and solar radiation pressure torques are proportional to the disk area and moment arm, and hence, scale with D^3, whereas the disk inertia scales with D^5.

To better understand the relative magnitudes of the angular accelerations, Fig. 3 plots Eqs. (7–10) as a function of the disk diameter at different orbit altitudes using the parameters listed in Table 1. These parameters model a thin disk with a constant density that matches the average spacecraft density from [28] and an offset between the center of pressure and the center of mass that is 1% of the diameter of the disk to account for manufacturing errors [26]. The figure plots α_{MD} for two different constant diameter growth rates: $\dot{D} = 10^{-1}$ m · s^{-1}, representative of a fast construction process analogous to the deployment of reflector antennas [5], and

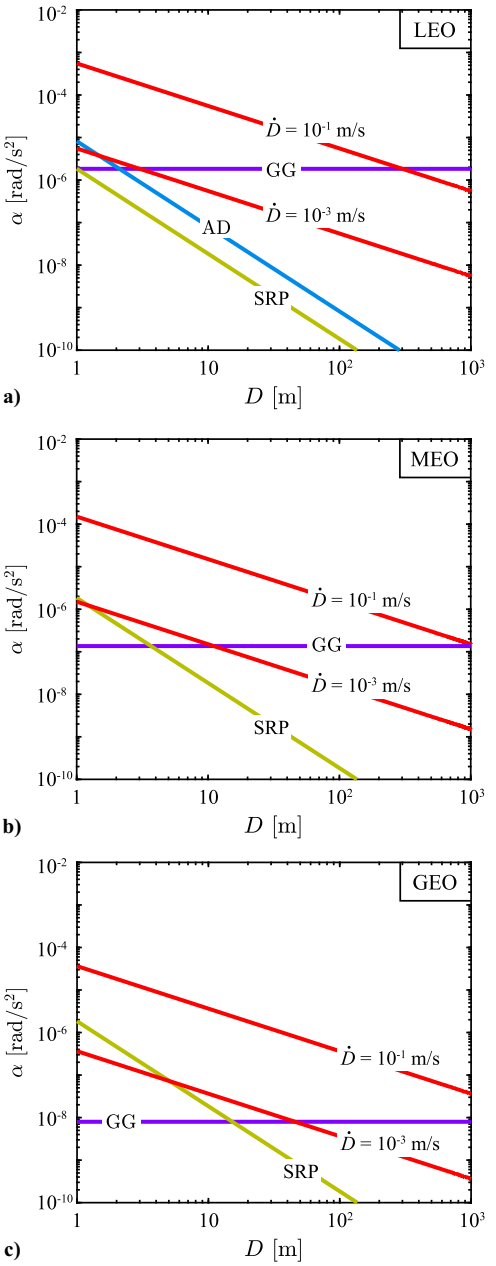


Fig. 3 Angular acceleration magnitudes as a function of disk diameter at different orbital altitudes: a) 500 km (LEO), b) 10,000 km (MEO), and c) 35,786 km (GEO).

Table 1 Disk and environmental disturbance parameters

Parameter	Value
ϵ	0.01
κ_T	0.01
ρ	79 kg · m^{-3} [28]
ρ_A (LEO)	6.97×10^{-13} kg · m^{-3} [29]
μ_E	3.986×10^{14} m^3 · s^{-2}
P	4.56×10^{-6} N · m^{-2}

$\dot{D} = 10^{-3}$ m · s^{-1}, representative of a slow construction process such as the extrusion of composite struts [10].

Figure 3 shows that for a fast diameter growth rate ($\dot{D} = 10^{-1}$ m · s^{-1}), the largest angular acceleration at all considered orbital altitudes and the majority of the considered length scales results from the rate of inertia change; i.e., the effects of environmental disturbances are negligible. However, for a slow diameter growth rate ($\dot{D} = 10^{-3}$ m · s^{-1}), the environmental disturbances are the dominant contributor to the attitude dynamics at some length scales. In low Earth orbit (LEO), gravity gradients dominate for larger diameters ($D > 5$ m). Likewise, in medium Earth orbit (MEO) and geosynchronous Earth orbit (GEO), solar radiation pressure or gravity gradients are dominant at most length scales. Environmental disturbances increasingly dominate the attitude dynamics as the growth rate further decreases. For context, $\dot{D} = 10^{-1}$ m · s^{-1} and $\dot{D} = 10^{-3}$ m · s^{-1} correspond to fabrication times for a 100-m diameter aperture on the order of 30 min and 1 day, respectively.

Taken together, these results demonstrate the importance of understanding the coupled effects of changes in mass properties and environmental disturbances on the attitude dynamics of ISAM spacecraft, particularly for large diameters ($D > 5$ m) and slow construction rates ($\dot{D} \leq 10^{-3}$ m · s^{-1}). They also emphasize that angular accelerations from gravity gradients are length-independent and the dominant environmental disturbance during ISAM across a wide range of length scales and orbital regimes. The effects of gravity gradients are further exacerbated for truss structures with sparsely distributed struts because their effective cross-sectional areas are substantially smaller than those of a solid disk, reducing the effects of the disturbance torques from atmospheric drag and solar radiation pressure considered above. Including the mass properties of the ISAM spacecraft in the analysis further decreases the relative magnitudes of the angular accelerations due to the time-varying mass distribution, atmospheric drag, and solar radiation pressure, whereas the angular acceleration due to gravity gradients does not change because it is independent of the mass properties. Hence, including the mass properties of the ISAM spacecraft has the effect of increasing the relative importance of gravity gradients on the attitude dynamics. For these reasons, the remainder of this paper focuses on the effects of gravity gradients during ISAM to develop the concept of passive gravity gradient capture.

III. Model for Gravity Gradient Capture During ISAM

This section uses the framework of variable-mass rigid-body dynamics [30,31] to develop an attitude dynamics model for ISAM that includes effects from both changes in the mass distribution during construction and gravity gradient torques. Variable-mass rigid-body dynamics extends classical rigid-body dynamics to systems that lose, gain, or redistribute mass.

Consider a control volume that consists of an ISAM spacecraft constructing a truss structure by redistributing mass from the spacecraft to the truss. For a circular orbit with mean motion $n = 2\pi/T$ and period T, the nondimensional equations of motion for this variable-mass system are [30]

$$\dot{\boldsymbol{q}} = \pi \boldsymbol{\Omega}(\hat{\boldsymbol{\omega}})\boldsymbol{q} \tag{11}$$

$$\dot{\hat{\boldsymbol{\omega}}} = -2\pi[\boldsymbol{I}_{\mathrm{CM}}]^{-1}(\hat{\boldsymbol{\omega}} \times [\boldsymbol{I}_{\mathrm{CM}}]\hat{\boldsymbol{\omega}}) + \hat{\boldsymbol{\alpha}}_{\mathrm{MD}} + \hat{\boldsymbol{\alpha}}_{\mathrm{GG}} \tag{12}$$

where dot notation now denotes derivatives with respect to nondimensional time $\hat{t} = t/T$. In Eq. (11), $\boldsymbol{q}$ is the unit quaternion describing the orientation of the spacecraft body frame with respect to the inertial frame and $\boldsymbol{\Omega}(\hat{\boldsymbol{\omega}})$ is [32]

$$\boldsymbol{\Omega}(\hat{\boldsymbol{\omega}}) \triangleq \begin{bmatrix} 0 & -\hat{\omega}_x & -\hat{\omega}_y & -\hat{\omega}_z \\ \hat{\omega}_x & 0 & \hat{\omega}_z & -\hat{\omega}_y \\ \hat{\omega}_y & -\hat{\omega}_z & 0 & \hat{\omega}_x \\ \hat{\omega}_z & \hat{\omega}_y & -\hat{\omega}_x & 0 \end{bmatrix} \tag{13}$$

The terms in Eq. (12) are expressed in the spacecraft body-fixed frame. Specifically, $\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}/n$ is the nondimensional angular velocity of the body frame with respect to the inertial frame and $[\boldsymbol{I}_{\mathrm{CM}}(\hat{t})]$ is the time-varying moment of inertia matrix about the instantaneous center of mass. Additionally, $\hat{\boldsymbol{\alpha}}_{\mathrm{MD}}$ and $\hat{\boldsymbol{\alpha}}_{\mathrm{GG}}$ are the nondimensional angular accelerations from the time-varying mass distribution and the gravity gradient torque given by [27,30]

$$\hat{\boldsymbol{\alpha}}_{\mathrm{MD}} = -[\boldsymbol{I}_{\mathrm{CM}}]^{-1}\left([\dot{\boldsymbol{I}}_{\mathrm{CM}}]\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \times \boldsymbol{h}_{\mathrm{CM}} + \frac{\dot{\boldsymbol{h}}_{\mathrm{CM}}}{2\pi}\right) \tag{14}$$

$$\hat{\boldsymbol{\alpha}}_{\mathrm{GG}} = 6\pi[\boldsymbol{I}_{\mathrm{CM}}]^{-1}(\hat{\boldsymbol{R}} \times [\boldsymbol{I}_{\mathrm{CM}}]\hat{\boldsymbol{R}}) \tag{15}$$

where

$$\boldsymbol{h}_{\mathrm{CM}} = \sum_{i=1}^{N} m_i(\boldsymbol{r}_i - \boldsymbol{r}_{\mathrm{CM}}) \times (\dot{\boldsymbol{r}}_i - \dot{\boldsymbol{r}}_{\mathrm{CM}}) \tag{16}$$

is the angular momentum of the bodies within the control volume relative to the instantaneous center of mass, and $\hat{\boldsymbol{R}}$ is the orbit radius unit vector pointing from the orbit center to the instantaneous center of mass, both measured relative to the origin of the spacecraft body frame. In Eq. (16), the angular momentum $\boldsymbol{h}_{\mathrm{CM}}$ is a function of the positions $\boldsymbol{r}_i$ of an arbitrary number of point masses m_i within the control volume and the position $\boldsymbol{r}_{\mathrm{CM}}$ of the instantaneous center of mass. In this paper, each point mass m_i represents a strut that moves relative to the spacecraft during the construction of a truss structure, as detailed in the Appendix. Together, Eqs. (11–16) govern the rigid-body attitude dynamics during ISAM with effects due to changes in mass distribution ($\hat{\boldsymbol{\alpha}}_{\mathrm{MD}}$) and gravity gradients ($\hat{\boldsymbol{\alpha}}_{\mathrm{GG}}$).

Gravity gradient capture occurs when the combined spacecraft and structure achieve a gravity-gradient-stabilized orientation after construction. There are multiple equilibrium orientations in which the principal inertia axes are aligned with the local-vertical, local-horizontal orbit frame such that there are no gravity gradient torques on a spacecraft [33]. In particular, gravity gradients provide a stabilizing torque when the minimum principal inertia axis is aligned with the orbit radius vector [34]. As a result, if an ISAM spacecraft is passively captured in this final orientation solely from changes in mass distribution and gravity gradients, active attitude control is unnecessary during construction.

To compute trajectories for passive gravity gradient capture, an important observation is that Eqs. (11–16) are invariant under the change of variables $(\hat{t}, \hat{\boldsymbol{\omega}}) \rightarrow (-\hat{t}, -\hat{\boldsymbol{\omega}})$; i.e., the trajectories are symmetric forward and backward in time. Thus, the equations of motion are *time reversible* [35]. Time reversibility provides a strategy for computing the initial conditions for trajectories that result in passive gravity gradient capture by solving Eqs. (11–16) backward in time from a gravity-gradient-stabilized final orientation.

IV. Gravity Gradient Capture During ISAM of Representative Structures

Using the time reversibility of Eqs. (11–16), this section illustrates the passive gravity gradient capture of ISAM spacecraft

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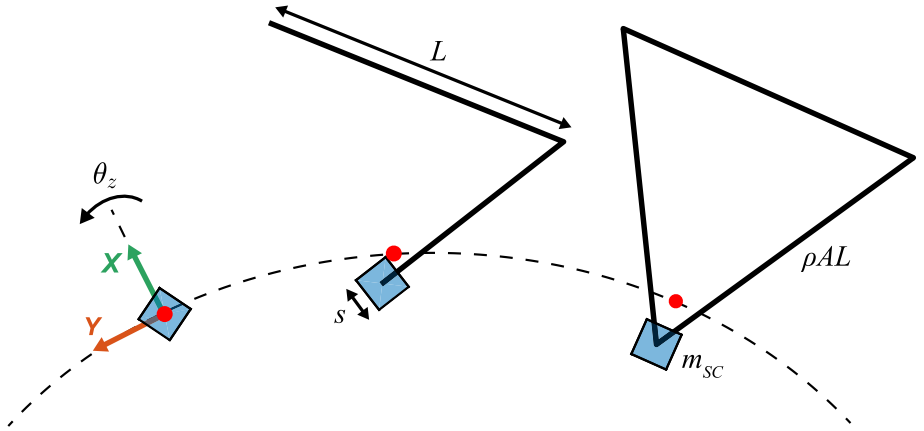


Fig. 4 Construction of a 2D triangle. Gravity gradient capture occurs when the triangle faces radially inward or outward, i.e., $\theta_z(\hat{t}_F) = 0$ or $\theta_z(\hat{t}_F) = \pi$.

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during the construction of two representative truss structures relevant for applications such as antennas and telescopes [36,37]: a 2D triangle unit cell (Sec. IV.A) and a 3D curved gridshell (Sec. IV.B).

A. Triangle Unit Cell

Consider the construction of a 2D equilateral triangle in the orbit plane (Fig. 4). The equilateral triangle may represent a unit cell for the 3D curved gridshell subsequently studied in Sec. IV.B. For this geometry, motion is restricted to the orbit plane, so the moment equilibrium of Eq. (12) simplifies to a single equation governing the dynamics of the pitch angle θ_z about the orbit-normal axis:

$$\ddot{\theta}_z = 3\pi\left(\frac{I_{xx} - I_{yy}}{I_{zz}}\right)\sin(2\theta_z) + 6\pi\left(\frac{I_{xy}}{I_{zz}}\right)\cos(2\theta_z) - \left(\frac{\dot{I}_{zz}}{I_{zz}}\right)(\dot{\theta}_z + 1) - \frac{1}{2\pi}\left(\frac{\dot{h}_z}{I_{zz}}\right) \tag{17}$$

Like Eq. (12), the dot notation in Eq. (17) denotes derivatives with respect to nondimensional time $\hat{t} = t/T$ (T is the orbit period), and $\dot{\theta}_z$ and $\ddot{\theta}_z$ are the angular velocity and acceleration normalized with respect to the orbit mean motion $n = 2\pi/T$ and n^2, respectively. The first two terms on the right-hand side of Eq. (17) are the angular accelerations from the gravity gradient torque, whereas the last two terms are the angular accelerations from the time-varying mass distribution. Here, I_{xx}, I_{yy}, and I_{xy} are the time-varying components of the moment of inertia matrix about the instantaneous center of mass,

$$[\boldsymbol{I}_{\mathrm{CM}}(\hat{t})] = \begin{bmatrix} I_{xx}(\hat{t}) & I_{xy}(\hat{t}) & \\ I_{xy}(\hat{t}) & I_{yy}(\hat{t}) & \\ & & I_{zz}(\hat{t}) \end{bmatrix} \tag{18}$$

and h_z is the nonzero component of the angular momentum from the motion of the struts and spacecraft relative to the instantaneous center of mass [cf. Eq. (16)]. Since there are no external forces other than gravity, the instantaneous center of mass remains on the circular orbit.

To compute the change in mass distribution during construction, the ISAM spacecraft sequentially extrudes struts, an approach that is representative of candidate ISAM processes for truss structures [8,9,14]. As described in the Appendix, the ISAM spacecraft is cubic with side length s and mass m_{SC} and constructs the triangle via the extrusion of three struts of length L and linear density ρA. Each strut is modeled as a slender rod of time-varying mass and length and is extruded according to the smooth profile used in [15] with zero velocity, acceleration, and jerk at the endpoints.

The extrusion of the three struts occurs over $\hat{t} \in [0, \hat{t}_F]$, where $\hat{t}_F$ is the construction timescale in number of orbits. After the extrusions, the final moments of inertia of the truss and spacecraft about their respective centers of mass are

$$[\boldsymbol{I}_T(\hat{t}_F)] = \frac{1}{4}\rho AL^3\,\mathrm{diag}\{1, 1, 2\} \tag{19}$$

$$[\boldsymbol{I}_{\mathrm{SC}}(\hat{t}_F)] = \frac{1}{6}m_{\mathrm{SC0}}s^2\boldsymbol{I}_{3\times3} \tag{20}$$

where $\boldsymbol{I}_{3\times3}$ is an identity matrix. For this specific geometry, Eq. (17) can be further simplified using the mass and inertia ratios between the truss and spacecraft after construction:

$$\eta_M \triangleq \frac{m_T(\hat{t}_F)}{m_{\mathrm{SC}}(\hat{t}_F)} = \frac{3\rho AL}{m_{\mathrm{SC0}}} \tag{21}$$

$$\eta_I \triangleq \frac{\|[\boldsymbol{I}_T(\hat{t}_F)]\|_2}{\|[\boldsymbol{I}_{\mathrm{SC}}(\hat{t}_F)]\|_2} = \left(\frac{3\rho AL}{m_{\mathrm{SC0}}}\right)\left(\frac{L}{s}\right)^2 \tag{22}$$

Once the mass and inertia ratios (η_M, η_I) and a construction timescale $(\hat{t}_F)$ are specified, Eq. (17) can be numerically integrated to compute the pitch angle θ_z as a function of time during construction.

Gravity gradient capture occurs when the triangle faces radially inward or outward, i.e., $\theta_z(\hat{t}_F) = 0$ or $\theta_z(\hat{t}_F) = \pi$. To compute passive trajectories that yield these stable final orientations, time reversibility can be used to solve Eq. (17) backward in time. Numerically, this involves integrating from some final time $\hat{t} = \hat{t}_F$ with specified final conditions to time $\hat{t} = 0$. Using this approach, Figs. 5 and 6 plot two representative passive trajectories for $(\eta_M, \eta_I, \hat{t}_F) = (0.03, 2.5, 1)$ and $(\eta_M, \eta_I, \hat{t}_F) = (0.06, 10, 1.5)$, respectively, which both result in the stable final orientation

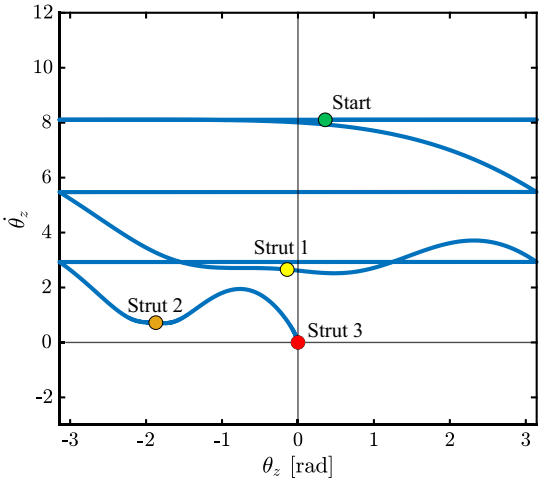


Fig. 5 Representative gravity gradient capture trajectory for the 2D triangle with $(\eta_M, \eta_I, \hat{t}_F) = (0.03, 2.5, 1)$ (Supplemental Video S1).

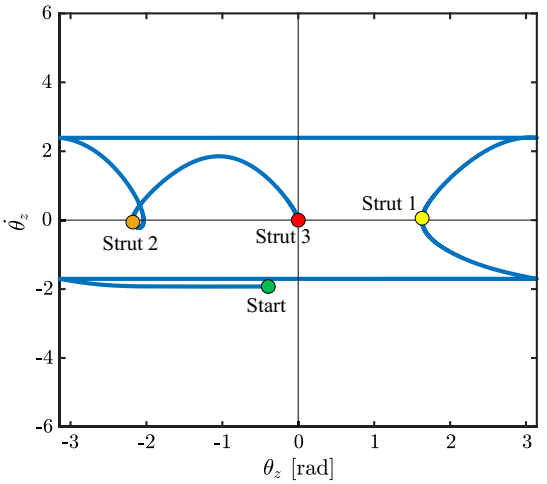


Fig. 6 Representative gravity gradient capture trajectory for the 2D triangle with $(\eta_M,\eta_I,\hat{t}_F) = (0.06,10,1.5)$ (Supplemental Video S2).

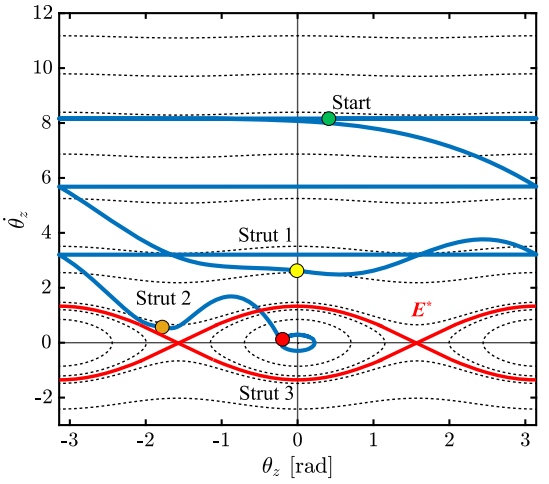


Fig. 7 Representative trajectory overlaid on energy isocontours (dotted). The critical energy E^* (red) separates stable oscillations from complete revolutions.

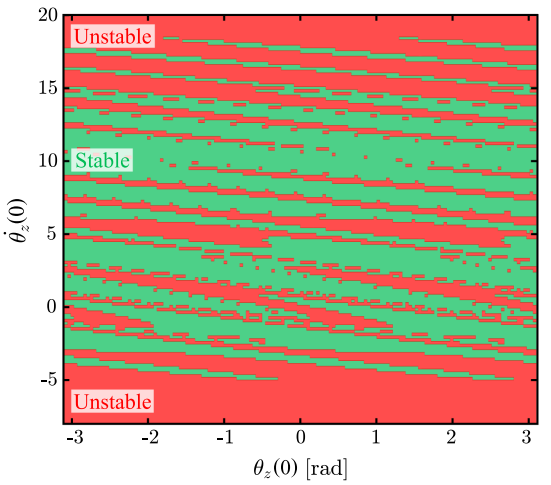


Fig. 8 Initial conditions for gravity gradient capture of the 2D triangle.

$\theta_z(\hat{t}_F) = 0$ (Supplemental Videos S1 and S2). The figures plot the trajectories in the phase plane $(\theta_z, \dot{\theta}_z)$ with the pitch rate nondimensionalized by the orbit mean motion n. The trajectory in Fig. 5 begins with a positive angular velocity that decreases to zero as the struts are extruded. Due to the high initial angular velocity of the spacecraft, the torque from the rate of change in inertia [the last two terms in Eq. (17)] dominates this trajectory. In contrast, the trajectory in Fig. 6 begins with a negative angular velocity and achieves a near zero angular velocity after the extrusion of each strut. The gravity gradient torque has a larger effect on this trajectory and causes the angular velocity to change signs multiple times. These qualitative differences between trajectories result from the different time-varying mass properties. For this 2D geometry, the trajectories can be classified into two categories: those with a positive angular velocity and those with an angular velocity that changes sign. These two types of trajectories demonstrate the feasibility of passive gravity gradient capture during the construction of the 2D triangle.

The trajectories in Figs. 5 and 6 achieve final equilibrium orientations with zero disturbance torques after the extrusion of the final strut. However, feasible trajectories for gravity gradient capture include any trajectory that results in a final state that remains stable in the presence of gravity gradients. For the 2D triangle, the total energy of the spacecraft after construction determines the stability of the final state. The total energy E is the sum of the rotational kinetic and gravitational potential energies [38]:

$$E(\theta_z,\dot{\theta}_z) = \frac{1}{2}I_{zz}(\hat{t}_F)(\dot{\theta}_z n)^2 + \frac{3}{4}n^2((I_{xx}(\hat{t}_F) - I_{yy}(\hat{t}_F))\cos(2\theta_z) - 2I_{xy}(\hat{t}_F)\sin(2\theta_z)) \quad (23)$$

In the absence of time-varying mass distribution, the gravity gradient torque results in attitude trajectories that conserve the total energy after construction, i.e., for $\hat{t} \geq \hat{t}_F$. Additionally, there is a critical energy E^*,

$$E^* = E(\pm\pi/2, 0) = -\frac{3}{4}n^2(I_{xx}(\hat{t}_F) - I_{yy}(\hat{t}_F)) \quad (24)$$

below which the spacecraft oscillates near the equilibrium points $(\theta,\dot{\theta}_z) = (0,0)$ or $(\theta,\dot{\theta}_z) = (\pi,0)$ and above which it tumbles, analogous to a simple pendulum. As a result, if $E(\hat{t}_F) < E^*$, the trajectory results in gravity gradient capture. To illustrate this, Fig. 7 plots a trajectory for the 2D triangle with a final state that is offset from the equilibrium point but with a subcritical total energy that results in stable oscillations.

The energy criterion facilitates the determination of the full range of initial conditions for passive gravity gradient capture of the 2D triangle. As an example, Fig. 8 depicts the range of initial conditions for $(\eta_M,\eta_I,\hat{t}_F) = (0.03, 2.5, 1)$ that result in stable final states. For high initial angular velocities ($\dot{\theta}_z(0) > 18.5$ and $\dot{\theta}_z(0) < -5$), Fig. 8 shows that any initial orientation results in unstable final states. However, within this range, there is a range of feasible initial conditions for gravity gradient capture that correspond to the green regions in the phase plane. The width of the regions varies between $\Delta\dot{\theta}_z(0) \approx [0.5, 2]$, with narrower regions for larger angular velocities. Here, the larger number of initial conditions with positive angular velocity is the result of the prograde circular orbit coupled with the fact that increasing inertia during construction decreases the angular velocity.

Although Fig. 8 highlights the feasibility of gravity gradient capture for the 2D triangle, an important practical consideration is the sensitivity to initial conditions. The effect of a small initial tilt angle ϕ out of the orbit plane on the trajectory for passive gravity gradient capture from Fig. 7 provides insight into this sensitivity. Figures 9 and 10 plot the trajectories for $(\eta_M,\eta_I,\hat{t}_F) = (0.03, 2.5, 1)$ with the initial tilt angles $\phi = 1$ deg and $\phi = 5$ deg. As the rotation of the spacecraft is no longer restricted to the orbit plane, these trajectories are solutions to the full 3D equations of motion [Eqs. (11–16)] and are plotted in the phase plane that consists of the three Euler angles and the three nondimensional angular velocity components in the body-fixed frame. When $\phi = 1$ deg, the trajectory in Fig. 9 is similar to the zero-tilt trajectory from Fig. 5, but with stable oscillations in roll and yaw. In contrast, when $\phi = 5$ deg, the initial angular velocity increases the tilt angle and results in an unstable final state that is far from the desired equilibrium point.

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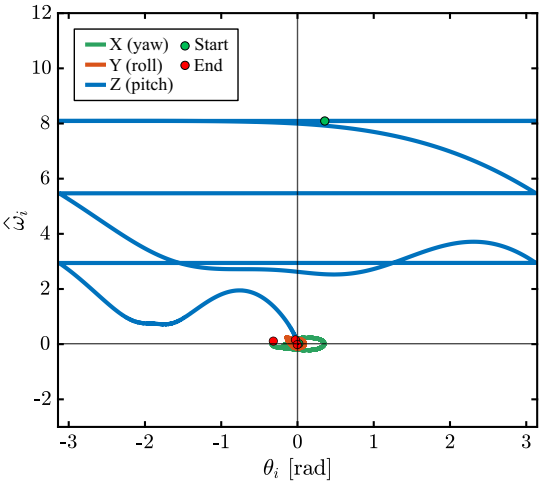


Fig. 9 Representative trajectory with $\phi = 1$ deg initial tilt.

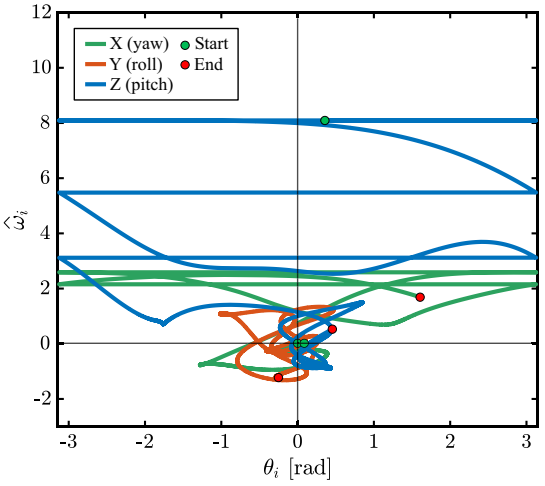


Fig. 10 Representative trajectory with $\phi = 5$ deg initial tilt.

To evaluate whether trajectories with an initial tilt result in gravity gradient capture, the full 3D equations of motion [Eqs. (11–16)] are linearized about the final state $(\boldsymbol{q}(\hat{t}_F), \hat{\boldsymbol{\omega}}(\hat{t}_F))$ and the eigenvalues of the corresponding Jacobian matrix are used to classify the stability of this final state [35]. Figures 11 and 12 use this linear stability

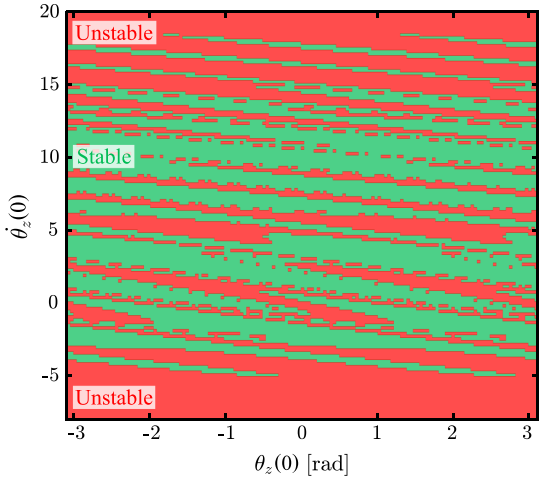


Fig. 11 Initial conditions for gravity gradient capture with $\phi = 1$ deg initial tilt.

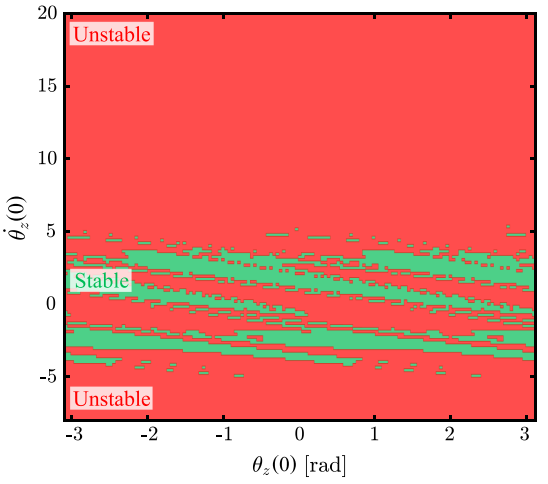


Fig. 12 Initial conditions for gravity gradient capture with $\phi = 5$ deg initial tilt.

criterion to plot the range of initial conditions that result in gravity gradient capture for $(\eta_M, \eta_I, \hat{t}_F) = (0.03, 2.5, 1)$ and the initial tilt angles $\phi = 1$ deg and $\phi = 5$ deg. The lower tilt angle $\phi = 1$ deg again has a small effect on the range of stable initial conditions (cf. Fig. 11); however, the higher tilt angle $\phi = 5$ deg destabilizes many trajectories, leaving stable initial conditions only within the narrow angular velocity range $\Delta\dot{\theta}_z(0) \approx [-5, 5]$ (cf. Fig. 12). Because the angular velocity is nondimensionalized with respect to the orbit mean motion, higher orbit altitudes are more sensitive to the initial tilt. Altogether, even though there is a wide range of initial conditions that result in passive gravity gradient capture for this 2D example, the stability of the final state is sensitive to initial condition perturbations.

B. Curved Gridshell

The second case study considers the more complex geometry of the 3D curved gridshell (Fig. 1) and uses the time reversibility of the 3D equations of motion [Eqs. (11–16)] to compute passive trajectories for gravity gradient capture, similar to the approach in 2D. The stable equilibrium orientation of the gridshell consists of the principal inertia axes aligned with the orbit frame such that $I_{xx} \leq I_{yy} \leq I_{zz}$ [33,34], as depicted in Fig. 1. The gridshell has diameter D, focal-length-to-diameter ratio F/D, average strut length L, strut linear density ρA, and total mass m_{T0} after construction, and the cubic spacecraft has side length s and mass m_{SC0} after construction. To compute the change in mass distribution, the spacecraft sequentially extrudes struts to construct the gridshell, as detailed in the Appendix, using a radially expanding construction sequence from the center of the gridshell, as described in [15].

With these assumptions, Fig. 13 plots a representative trajectory for passive gravity gradient capture of a curved gridshell over the time horizon $\hat{t}_F = 1$ with the parameters from Table 2 (Supplemental Video S3). This trajectory is more complex than the trajectories for the 2D triangle and exhibits tumbling about all three body axes up until the stable final orientation is reached at the end of construction. For this example, the mass and inertia ratios between the spacecraft and gridshell after construction are $\eta_M \approx 0.28$ and $\eta_I \approx 28$, much higher than those for the 2D triangle, resulting in an almost order-of-magnitude higher initial angular velocity. High initial angular velocities may pose practical challenges in implementing gravity gradient capture; for example, they may be incompatible with angular rate requirements for sensitive instruments such as star trackers [39] or cause large-amplitude vibrations from high inertial loads. Section V discusses strategies to address these issues and enable robust gravity gradient capture of large structures.

In 3D, the total energy of the spacecraft is again conserved after construction is complete [40], but the three orientation parameters

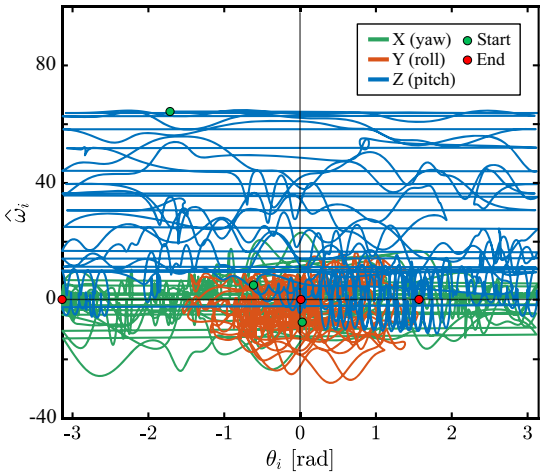


Fig. 13 Representative gravity gradient capture trajectory for the 3D gridshell (Supplemental Video S3).

Table 2 Gridshell and spacecraft parameters

Parameter	Value
D	20 m
F/D	0.5
L/D	0.1
ρA	0.25 kg/m
s	2 m
m_{T0}	140 kg
m_{SC0}	500 kg

complicate the energy contours near the equilibrium points and make it difficult to determine the range of initial conditions for gravity gradient capture. However, as in the 2D case study, the sensitivity to initial conditions is of practical importance. The remainder of this section uses a Monte Carlo analysis that randomizes the initial conditions corresponding to the trajectory in Fig. 13 to assess this sensitivity. The analysis then determines the maximum perturbations to the initial orientation and angular velocity, denoted by $\delta\theta^*$ and $\delta\hat{\omega}^*$, for which 90% of the sampled trajectories result in gravity gradient capture. In this case, gravity gradient capture is evaluated using the linear stability criterion for the full 3D equations of motion [Eqs. (11–16)].

For simplicity, the Monte Carlo simulations consider random perturbations to either the initial orientation or the initial angular velocity, not both. To generate uniform random initial orientation perturbations, each simulation samples a unit quaternion $\delta\boldsymbol{q}$ that represents a uniformly distributed rotation $\theta = U(0, \delta\theta)$ about a uniform random unit vector $\hat{\boldsymbol{e}}_1 = [\hat{e}_x, \hat{e}_y, \hat{e}_z]^T$ [41]; i.e.,

$$\delta\boldsymbol{q} = \left[\cos\frac{\theta}{2}, \hat{e}_x \sin\frac{\theta}{2}, \hat{e}_y \sin\frac{\theta}{2}, \hat{e}_z \sin\frac{\theta}{2}\right]^T \tag{25}$$

Similarly, to generate initial angular velocity perturbations, each simulation samples an angular velocity with uniformly distributed norm along a different uniform random unit vector $\hat{\boldsymbol{e}}_2$; i.e.,

$$\delta\hat{\boldsymbol{\omega}} = U(0, \delta\hat{\omega})\hat{\boldsymbol{e}}_2 \tag{26}$$

Figure 14 plots the results of Monte Carlo analyses with 5000 randomly sampled perturbations applied to the trajectory in Fig. 13. Specifically, the figure depicts histograms of $\delta\theta$ and $\delta\hat{\omega}$ that result in gravity gradient capture. In this example, the maximum orientation and angular velocity perturbations that still achieve gravity gradient capture are $\delta\theta^* \approx 7.5 \times 10^{-3}$ rad and $\delta\hat{\omega}^* \approx 0.05$, respectively. These correspond to stringent attitude control requirements relative to the initial angular velocity for the trajectory in Fig. 13 ($\|\hat{\boldsymbol{\omega}}(0)\|_2 \approx 65$). As in 2D, the angular velocity sensitivity increases for higher orbital altitudes because it is normalized with respect to the orbit mean motion.

Although this Monte Carlo analysis is specific to the parameters in Table 2, Monte Carlo simulations for different gridshell and spacecraft parameters yielded similar results. In particular, numerical experiments showed that the sensitivity to initial conditions increased for both larger final mass/inertia ratios and longer construction timescales. In addition, numerical experiments for other 3D truss geometries (including the 3D tetrahedral truss from [37]) exhibited similar sensitivities. The next section discusses strategies for decreasing these sensitivities.

V. Strategies for Robust Gravity Gradient Capture

The case studies in Sec. IV highlight the feasibility of gravity gradient capture during ISAM, but also reveal practical challenges related to high initial angular velocities and sensitivities to initial conditions. This section discusses alternate strategies and potential solutions.

The first strategy to decrease the initial angular velocity is to implement gravity gradient capture in stages, as was done on the RAE spacecraft. The RAE spacecraft specifically used gravity gradient capture three times during boom deployment: first at 112 m, then at 138 m, and finally at over 200 m [25]. The analogous approach for ISAM would segment a large structure into subassemblies and use passive gravity gradient capture to stabilize the orientation after the construction of each subassembly. In this way,

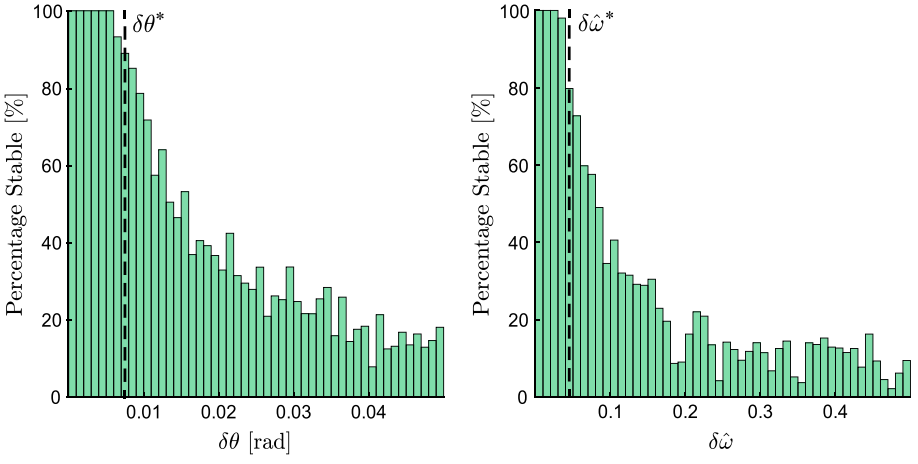


Fig. 14 Histograms of initial perturbations to the Fig. 13 trajectory that achieve gravity gradient capture.

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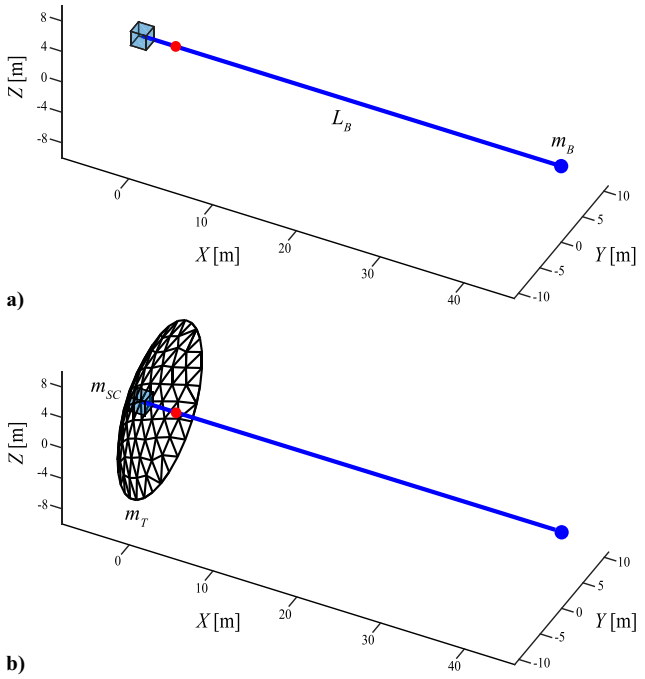


Fig. 15 a) Boom deployment along the orbit radius vector, followed by b) construction of the gridshell.

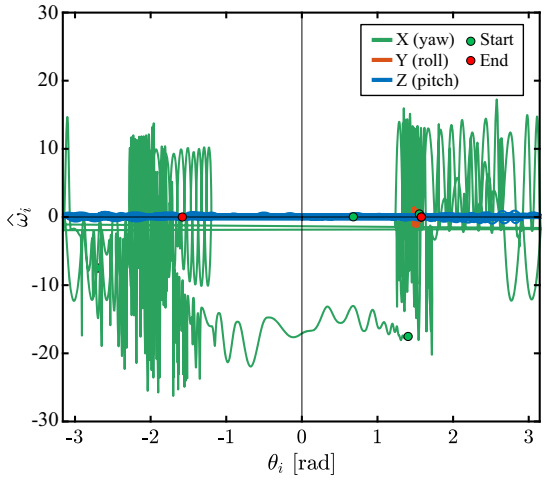


Fig. 16 Representative gravity gradient capture trajectory after boom deployment (Supplemental Video S4).

each stage has a smaller inertia change and initial angular velocity as compared to construction of the entire structure in a single stage. In addition, planning the construction process such that the largest inertia change occurs in the first stage, e.g., by constructing the perimeter of the structure first, would significantly decrease the initial angular velocities required for gravity gradient capture in subsequent stages.

Another approach to decrease the initial angular velocity is to deploy a boom along the orbit radius vector before construction. Intuitively, the additional boom inertia helps stabilize the trajectory by maintaining the minimum principal inertia axis along the orbit radius vector. As an example, consider a gravity gradient boom deployed along the gridshell axis such that the final stable orientation consists of the gridshell facing either toward or away from Earth, as depicted in Fig. 15. The presence of the boom significantly decreases the initial angular velocity required for passive gravity gradient capture of the gridshell. As an example, Fig. 16 depicts a representative trajectory for passive gravity gradient capture using the parameters in Table 2 and a boom of length $L_B = 50$ m, tip mass $m_{\text{tip}} = 50$ kg, and boom density equal to the strut density (Supplemental Video S4). This trajectory has an initial angular velocity with a magnitude three times smaller than that of the trajectory sans boom (cf. Fig. 13). Repeating the Monte Carlo analyses from Sec. IV.B with the boom shows that the boom also significantly decreases the initial condition sensitivity. In particular, the results in Fig. 17 show that the maximum initial orientation and angular velocity perturbations that still achieve gravity gradient capture are $\delta\theta^* \approx 0.06$ rad and $\|\delta\hat{\boldsymbol{\omega}}\|_2^* \approx 1.05$; i.e., they are an order of magnitude larger than those sans boom. Thus, the use of a gravity gradient boom can result in lower initial angular velocities and more robust trajectories. More generally, designing the construction process to maintain the minimum principal inertia axis in the direction of the orbit radius vector can help facilitate the robust gravity gradient capture of large structures.

A final strategy involves using the construction rate of the structure itself to actively control the spacecraft's attitude and guide it to gravity gradient capture. Previous work has investigated using changes in mass properties as a method of attitude control [42,43], e.g., to flip a spacecraft using the intermediate-axis instability [43], and similar strategies may be useful for efficient attitude control during ISAM.

VI. Conclusions

In-space assembly and manufacturing (ISAM) pose challenges to attitude control system design due to the varying mass properties and environmental disturbances during construction. This paper has described a passive approach to attitude control during ISAM, referred to as passive gravity gradient capture, that combines the

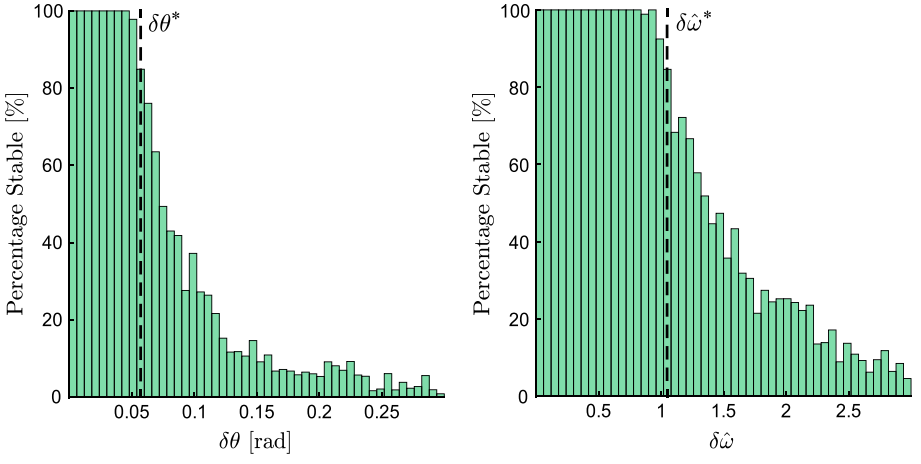


Fig. 17 Histograms of initial perturbations to the Fig. 16 trajectory that achieve gravity gradient capture.

effects of changes in mass properties and gravity gradients to achieve a stable final orientation at the end of construction. The main benefit of this approach is that it eliminates any requirement for active attitude control during construction. As a result, it removes constraints on the fabrication time of large structures associated with attitude control system capabilities. Two case studies, focused on the construction of 2D and 3D truss structures in circular Earth orbit, demonstrated that trajectories for passive gravity gradient capture can be computed for ISAM spacecraft undergoing complex inertia changes. These case studies also identified practical challenges associated with high initial angular velocities and initial condition sensitivities. However, strategies such as deploying a gravity gradient boom before construction were found to mitigate these challenges.

The results in this paper provide a starting point for future work on strategies that exploit in-space environmental disturbances for attitude control during ISAM. Besides gravity gradients, atmospheric drag and magnetic fields may be used for attitude control in low Earth orbit, and solar radiation pressure may be used for attitude control in geosynchronous Earth orbit and deep space. Thoughtful integration of these environmental effects into the design of the construction process may enable efficient attitude control and fast fabrication times for future ISAM missions.

Appendix: Sequential Strut Extrusion

The case studies in this paper assume a construction process for truss structures that relies on the sequential extrusion of struts. This appendix describes the modeling approach for sequential strut extrusion.

As an example, consider a cubic spacecraft extruding a strut that is connected to three previously extruded struts (Fig. A1). The spacecraft extrudes the new strut in the direction $\boldsymbol{n}_{x1}$, and the strut has length $\Gamma(t)$ during the extrusion period $t \in [0, \Delta t]$. In this paper, $\Gamma(t)$ follows the smooth extrusion profile with zero velocity, acceleration, and jerk at the endpoints used in [15]. During each extrusion, mass moves from the spacecraft to the extruding strut such that the combined system mass M remains constant. The instantaneous mass m_T of the truss and m_{SC} of the spacecraft are given by

$$m_T(t) = \sum_{i=1}^{N} m_i(t) \tag{A1}$$

$$m_{\mathrm{SC}}(t) = M - m_T(t) \tag{A2}$$

where $m_i(t)$ is the mass of strut i (e.g., $i = 1$ corresponds to the extruding strut), and N is the total number of struts in the current geometry (e.g., $N = 4$ in Fig. A1). If each strut has a constant linear density ρA and mass only transfers from the spacecraft to the extruding strut, then

$$m_i(t) = \begin{cases} \rho A \Gamma(t), & i = 1, \\ \rho A L_i, & 2 \le i \le N \end{cases} \tag{A3}$$

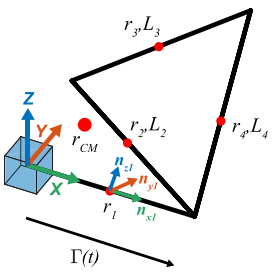


Fig. A1 Sequential strut extrusion with extrusion profile $\Gamma(t)$.

where L_i is the length of the previously extruded strut i. During extrusion, the location of the instantaneous center of mass $\boldsymbol{r}_{\mathrm{CM}}(t)$ of the combined system varies relative to the body-fixed frame at the center of mass of the spacecraft and is given by

$$\boldsymbol{r}_{\mathrm{CM}}(t) = \frac{1}{M} \sum_{i=1}^{N} m_i(t) \boldsymbol{r}_i(t) \tag{A4}$$

where $\boldsymbol{r}_i(t)$ is the time-varying center of mass of strut i relative to the spacecraft body-fixed frame:

$$\boldsymbol{r}_i(t) = \begin{cases} \frac{1}{2}\Gamma(t)\boldsymbol{n}_{x1}, & i = 1, \\ \boldsymbol{r}_{i0} + \Gamma(t)\boldsymbol{n}_{x1}, & 2 \le i \le N \end{cases} \tag{A5}$$

In Eq. (A5), $\boldsymbol{r}_{i0}$ denotes the center of mass of previously extruded strut $i > 1$ at the beginning of extrusion of strut $i = 1$.

The instantaneous mass moments of inertia relative to the instantaneous center of mass of the combined system during the extrusion are

$$[\boldsymbol{I}_{\mathrm{CM}}(t)] = [\boldsymbol{I}_{\mathrm{SC}}(t)] + [\boldsymbol{I}_T(t)] + M[\boldsymbol{r}_{\mathrm{CM}}(t)]_\times [\boldsymbol{r}_{\mathrm{CM}}(t)]_\times \tag{A6}$$

where $[\boldsymbol{I}_{\mathrm{SC}}(t)]$ and $[\boldsymbol{I}_T(t)]$ are the mass moments of inertia of the spacecraft and the truss relative to the spacecraft body-fixed frame, respectively, and $[\boldsymbol{x}]_\times$ denotes the skew-symmetric matrix for a vector $x \in \mathbb{R}^3$:

$$[\boldsymbol{x}]_\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \tag{A7}$$

For a spacecraft modeled as a cube with side length s,

$$[\boldsymbol{I}_{\mathrm{SC}}(t)] = \frac{1}{6} m_{\mathrm{SC}}(t) s^2 \boldsymbol{I}_{3\times 3} \tag{A8}$$

where $\boldsymbol{I}_{3\times 3}$ is an identity matrix. Similarly, the moments of inertia of the truss are

$$[\boldsymbol{I}_T(t)] = \sum_{i=1}^{N} ([\boldsymbol{R}_i][\boldsymbol{I}_i(t)][\boldsymbol{R}_i]^T - m_i(t)[\boldsymbol{r}_i(t)]_\times [\boldsymbol{r}_i(t)]_\times) \tag{A9}$$

where $[\boldsymbol{I}_i(t)]$ is the mass moment of inertia matrix of strut i about its center of mass in the local strut frame. In Eq. (A9), the matrix $[\boldsymbol{R}_i]$ describes the transformation $[\boldsymbol{R}_i] = \{\boldsymbol{n}_{xi}, \boldsymbol{n}_{yi}, \boldsymbol{n}_{zi}\}$ from the spacecraft body-fixed frame to the local strut frame (aligned such that $\boldsymbol{n}_{xi}$ is parallel to the axis of the strut). Modeling each strut as a thin, slender rod then provides the following expression for $[\boldsymbol{I}_i(t)]$:

$$[\boldsymbol{I}_i(t)] = \begin{cases} \frac{1}{12}\rho A \Gamma(t)^3 \,\mathrm{diag}\{0, 1, 1\}, & i = 1, \\ \frac{1}{12} m_i L_i^2 \,\mathrm{diag}\{0, 1, 1\}. & 2 \le i \le N \end{cases} \tag{A10}$$

Equations (A1–A10) describe the variation in the mass properties of the combined spacecraft and truss system during the extrusion of a new strut. After each extrusion, the total number of struts N increments by one, and the extrusion process repeats until the entire geometry is constructed. Following our previous work [14], the construction sequence is an Euler path through the truss geometry that dictates the extrusion directions $\boldsymbol{n}_{xi}$, the strut lengths L_i, and the total number of struts N. The resulting time-varying mass properties during construction are used in the variable-mass rigid-body dynamics model of Sec. III.

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